CN-SDN MININET

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Controller setup

```
varun@varun-VirtualBox:~/pox$ cd ~/pox
varun@varun-VirtualBox:~/pox$ ./pox.py log.level --DEBUG openflow.of_01 qos_cont
roller
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
DEBUG:core:POX 0.7.0 (gar) going up...
DEBUG:core:Running on CPython (3.8.10/Mar 18 2025 20:04:55)
DEBUG:core:Platform is Linux-5.15.0-139-generic-x86_64-with-glibc2.29
WARNING:version:Support for Python 3 is experimental.
INFO:core:POX 0.7.0 (gar) is up.
DEBUG:openflow.of_01:Listening on 0.0.0.0:6633
```

Controller-switch interaction:

```
varun@varun-VirtualBox: ~
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
varun@varun-VirtualBox:~$ sudo mn --topo single,3 --controller=remote,ip=127.0.0
.1,port=6633
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>
```

```
varun@varun-VirtualBox:~/pox$ cd ~/pox
varun@varun-VirtualBox:~/pox$ ./pox.py forwarding.l2_learning
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
WARNING:version:Support for Python 3 is experimental.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 2] connected
```

Ping output + ICMP logs

```
varun@varun-VirtualBox: ~
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=28.8 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=11.5 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=17.6 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=34.2 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=9.01 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4010ms
rtt min/avg/max/mdev = 9.012/20.220/34.201/9.779 ms
mininet>
```

```
varun@varun-VirtualBox: ~/pox
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
DEBUG:core:POX 0.7.0 (gar) going up...
DEBUG:core:Running on CPython (3.8.10/Mar 18 2025 20:04:55)
DEBUG:core:Platform is Linux-5.15.0-139-generic-x86_64-with-glibc2.29
WARNING:version:Support for Python 3 is experimental.
INFO:core:POX 0.7.0 (gar) is up.
DEBUG:openflow.of_01:Listening on 0.0.0.0:6633
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:qos_controller:Switch connected
INFO:openflow.of_01:[00-00-00-00-00-01 1] closed
INFO:openflow.of_01:[00-00-00-00-00-01 3] connected
INFO:qos_controller:Switch connected
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
```

iperf output + TCP logs :

```
.1,port=6633
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> h2 iperf -s
------------------------------------------------------------
Server listening on TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
^Cmininet> h1 iperf -c h2
connect failed: Connection refused
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> h2 iperf -s &
mininet> h2 ps
------------------------------------------------------------
Server listening on TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
    PID TTY          TIME CMD
   5941 pts/5    00:00:00 bash
   6018 pts/5    00:00:00 iperf
   6021 pts/5    00:00:00 ps
mininet> h1 iperf -c h2
------------------------------------------------------------
Client connecting to 10.0.0.2, TCP port 5001
TCP window size: 1.05 MByte (default)
------------------------------------------------------------
[  3] local 10.0.0.1 port 43178 connected with 10.0.0.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  3]  0.0-10.0 sec  22.3 GBytes  19.1 Gbits/sec
```

Flow table:

```
mininet> dpctl dump-flows
*** s1 ------------------------------------------------------------------------
 cookie=0x0, duration=7.913s, table=0, n_packets=5, n_bytes=490, idle_timeout=10, hard_timeout=30
, priority=65535,icmp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=f2:9d:54:34:2e:7d,dl_dst=a2:0a:bf:
83:48:e0,nw_src=10.0.0.1,nw_dst=10.0.0.2,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth2
"
 cookie=0x0, duration=7.919s, table=0, n_packets=5, n_bytes=490, idle_timeout=10, hard_timeout=30
, priority=65535,icmp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=a2:0a:bf:83:48:e0,dl_dst=f2:9d:54:
34:2e:7d,nw_src=10.0.0.2,nw_dst=10.0.0.1,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth1
"
 cookie=0x0, duration=2.750s, table=0, n_packets=2, n_bytes=84, idle_timeout=10, hard_timeout=30,
 priority=65535,arp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=f2:9d:54:34:2e:7d,dl_dst=a2:0a:bf:83
:48:e0,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=1 actions=output:"s1-eth2"
 cookie=0x0, duration=2.749s, table=0, n_packets=2, n_bytes=84, idle_timeout=10, hard_timeout=30,
 priority=65535,arp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=a2:0a:bf:83:48:e0,dl_dst=f2:9d:54:34
:2e:7d,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=1 actions=output:"s1-eth1"
 cookie=0x0, duration=2.746s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30,
 priority=65535,arp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=a2:0a:bf:83:48:e0,dl_dst=f2:9d:54:34
:2e:7d,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=2 actions=output:"s1-eth1"
 cookie=0x0, duration=2.746s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30,
 priority=65535,arp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=f2:9d:54:34:2e:7d,dl_dst=a2:0a:bf:83
:48:e0,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=2 actions=output:"s1-eth2"
```

Scenario 2:

Blocked traffic:

```
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=35.8 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=14.0 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=9.90 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=11.7 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=12.5 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4013ms
rtt min/avg/max/mdev = 9.901/16.800/35.846/9.615 ms
mininet> h1 ping -c 5 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.

--- 10.0.0.3 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4090ms

mininet> dpctl dump-flows
*** s1 ------------------------------------------------------------------------
mininet>
```

```
varun@varun-VirtualBox: ~/pox
INFO:core:POX 0.7.0 (gar) is up.
DEBUG:openflow.of_01:Listening on 0.0.0.0:6633
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:qos_controller:Switch connected
INFO:openflow.of_01:[00-00-00-00-00-01 1] closed
INFO:openflow.of_01:[00-00-00-00-00-01 3] connected
INFO:qos_controller:Switch connected
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:HIGH PRIORITY: ICMP packet
INFO:qos_controller:BLOCKED: h1 -> h3 traffic
INFO:qos_controller:BLOCKED: h1 -> h3 traffic
INFO:qos_controller:BLOCKED: h1 -> h3 traffic
INFO:qos_controller:BLOCKED: h1 -> h3 traffic
```

Flow table after blocked traffic:

```
mininet> dpctl dump-flows
*** s1 ------------------------------------------------------------------------
 cookie=0x0, duration=8.910s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30,
 priority=65535,arp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=ba:6b:b6:a8:01:85,dl_dst=0e:43:a6:13
:5f:54,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=1 actions=output:"s1-eth1"
 cookie=0x0, duration=8.903s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30,
 priority=65535,arp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=0e:43:a6:13:5f:54,dl_dst=ba:6b:b6:a8
:01:85,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=2 actions=output:"s1-eth2"
 cookie=0x0, duration=6.693s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30,
 priority=65535,arp,in_port="s1-eth3",vlan_tci=0x0000,dl_src=ea:36:cf:8d:22:c8,dl_dst=0e:43:a6:13
:5f:54,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,arp_op=2 actions=output:"s1-eth1"
 cookie=0x0, duration=1.495s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30,
 priority=65535,arp,in_port="s1-eth3",vlan_tci=0x0000,dl_src=ea:36:cf:8d:22:c8,dl_dst=0e:43:a6:13
:5f:54,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,arp_op=1 actions=output:"s1-eth1"
 cookie=0x0, duration=1.481s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30,
 priority=65535,arp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=0e:43:a6:13:5f:54,dl_dst=ea:36:cf:8d
:22:c8,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,arp_op=2 actions=output:"s1-eth3"
 cookie=0x0, duration=14.144s, table=0, n_packets=5, n_bytes=490, idle_timeout=10, hard_timeout=3
0, priority=65535,icmp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=0e:43:a6:13:5f:54,dl_dst=ba:6b:b6
:a8:01:85,nw_src=10.0.0.1,nw_dst=10.0.0.2,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth
2"
 cookie=0x0, duration=14.138s, table=0, n_packets=5, n_bytes=490, idle_timeout=10, hard_timeout=3
0, priority=65535,icmp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=ba:6b:b6:a8:01:85,dl_dst=0e:43:a6
:13:5f:54,nw_src=10.0.0.2,nw_dst=10.0.0.1,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth
1"
 cookie=0x0, duration=6.690s, table=0, n_packets=5, n_bytes=490, idle_timeout=10, hard_timeout=30
, priority=65535,icmp,in_port="s1-eth1",vlan_tci=0x0000,dl_src=0e:43:a6:13:5f:54,dl_dst=ea:36:cf:
8d:22:c8,nw_src=10.0.0.1,nw_dst=10.0.0.3,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth3
"
 cookie=0x0, duration=6.684s, table=0, n_packets=5, n_bytes=490, idle_timeout=10, hard_timeout=30
, priority=65535,icmp,in_port="s1-eth3",vlan_tci=0x0000,dl_src=ea:36:cf:8d:22:c8,dl_dst=0e:43:a6:
13:5f:54,nw_src=10.0.0.3,nw_dst=10.0.0.1,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth1
"
```

SDN Mininet Project Report

1. Introduction

Software Defined Networking (SDN) separates the control plane from the data plane, allowing centralized control of network behavior through a controller. This project demonstrates SDN concepts using Mininet and the POX controller, focusing on controller–switch interaction, flow rule installation, and performance observation.

---

2. Objective

- Set up an SDN network using Mininet
- Establish controller–switch communication
- Observe flow rule (match–action) behavior
- Analyze network performance using ping and iperf

---

3. Tools Used

- Mininet (network emulation)
- POX Controller (OpenFlow controller)
- Open vSwitch (virtual switch)
- ping (latency testing)
- iperf (throughput testing)

---

4. Topology

A simple topology was created:

- 1 switch (s1)
- 3 hosts (h1, h2, h3)
- Remote POX controller

---

5. Implementation

The switch successfully connected to the controller, establishing communication required for SDN operation.

---

6. Testing & Results

a) Connectivity Test (Ping)

h1 ping -c 5 h2

- Result: 0% packet loss
- Confirms proper connectivity and packet forwarding

---

b) Throughput Test (iperf)

h2 iperf -s
h1 iperf -c h2

- Result: High throughput observed (Gbps range)
- Confirms successful TCP communication

---

c) Flow Table Observation

dpctl dump-flows

- Flow entries appeared after traffic generation
- Demonstrates dynamic rule installation (match–action logic)

---

7. Analysis

- Controller dynamically installs flow rules based on traffic
- Ping shows low latency communication
- iperf shows high throughput due to virtual environment
- Demonstrates SDN principles: centralized control and programmable network

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8. Conclusion

The project successfully demonstrates SDN functionality using Mininet and POX. Controller–switch interaction, flow rule installation, and performance metrics were observed and validated. The experiment provides a clear understanding of SDN behavior in a virtual network.

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9. Future Scope

- Implement QoS-based traffic prioritization
- Use advanced controllers like Ryu
- Extend to larger network topologies